HitPredict: Predicting Billboard Hits Using Spotify Audio Features

UE24CS352A Machine Learning — Mini-Project, Problem 21 | Team: Shahid, Vilas

1. Problem Statement

Predict whether a song will become a Billboard Hot 100 hit using measurable properties of the song and its artist, rather than subjective judgment. This is a binary classification problem: given a song's Spotify audio features (and the artist's chart history), output hit (1) or non-hit (0), with a probability score. The project is based on the Stanford CS229 2018 report 'HitPredict: Predicting Billboard Hits Using Spotify Data.'

2. Dataset

Two public Kaggle datasets were merged: the **Spotify Tracks Dataset** (~114K tracks with 13 audio features — danceability, energy, loudness, speechiness, acousticness, instrumentalness, liveness, valence, tempo, duration, key, mode, time signature) and the **Billboard Hot 100 dataset** (all chart entries, 1958–2021). Songs were matched on cleaned title and primary artist name (lowercased, punctuation and 'feat.' text stripped). A song was labeled 1 if it appeared on the Billboard chart between 1990–2018, else 0. The matched set was balanced 50/50 (hits vs. an equal random sample of non-hits) for model comparison, and also evaluated unbalanced at the dataset's true hit rate of 2.08%, to test realistic performance.

3. Approach

Baseline: trained five classifiers — Logistic Regression, SVM (RBF), Random Forest, Gradient Boosting, and a small MLP — on the 13 audio features, with a stratified 75/25 train/test split. The *popularity* column was deliberately excluded as a predictor, since it is a consequence of being a hit rather than an independent signal (label leakage).

Artist history feature: following the original paper's finding that artist history was the strongest predictor, we engineered *artist_hits* — the count of the artist's other Billboard hits in the same window, excluding the song itself to avoid leakage. This feature alone accounted for 57% of Random Forest's feature importance after it was added.

Tuning: RandomizedSearchCV (5-fold CV, ROC-AUC scoring) over Random Forest and SVM hyperparameters to reduce the gap between train and test accuracy.

Imbalanced evaluation: re-tested the tuned models on the full, realistic 2.08%-hit-rate dataset using class-weighted training, and judged by precision/recall/F1/PR-AUC rather than accuracy, since a model predicting 'non-hit' always would score ~98% accuracy.

4. Implementation Overview

Pipeline: **build_dataset.py** (merge & label) → **add_artist_score.py** (artist history feature) → **train.py** (baseline models) → **tune.py** (hyperparameter search + imbalanced evaluation) → **demo.py** (CLI: predicts hit probability for a song, by lookup or manual feature entry). Built in Python with pandas and scikit-learn. All code is in the GitHub repository with a README describing setup and run order.

5. Results

Model / Setting	Test Acc	Precision	Recall	ROC-AUC
RF, audio features only (balanced)	0.740	0.715	0.798	0.822
RF, + artist_hits, tuned (balanced)	0.912	0.939	0.881	0.958
RF, + artist_hits, class-weighted (realistic, 2.08% hit rate)	0.948	0.262	0.840	0.955

Adding artist_hits raised ROC-AUC from 0.822 to 0.958, confirming the original paper's conclusion that artist chart history outweighs any single audio feature. On the realistic imbalanced data, the class-weighted Random Forest catches 84% of true hits at 26.2% precision — about 13x better than the 2.08% base rate, making it a useful screening tool rather than a high-confidence oracle.

6. Conclusions

Audio features alone are a weak predictor of commercial success (ROC-AUC 0.82); who is performing the song matters far more than how it sounds. A simple demo illustrates this directly: a cover of 'Shape of You' by an unknown artist was misclassified as a likely hit using audio features alone (62% hit probability) but correctly classified as unlikely (22%) once artist history was added, while the original Ed Sheeran version's predicted probability rose from 49% to 90%. Limitations: non-hit songs were sampled across all eras and genres while hits were restricted to 1990–2018 mainstream chart songs, so some features may partly separate 'mainstream pop' from 'everything else' rather than hit from non-hit; artist_hits counts chart appearances in the same window rather than strictly before each song's release, since exact release dates were not available. Future work could incorporate true release-date ordering, lyrics/sentiment features, and streaming/playlist placement data.